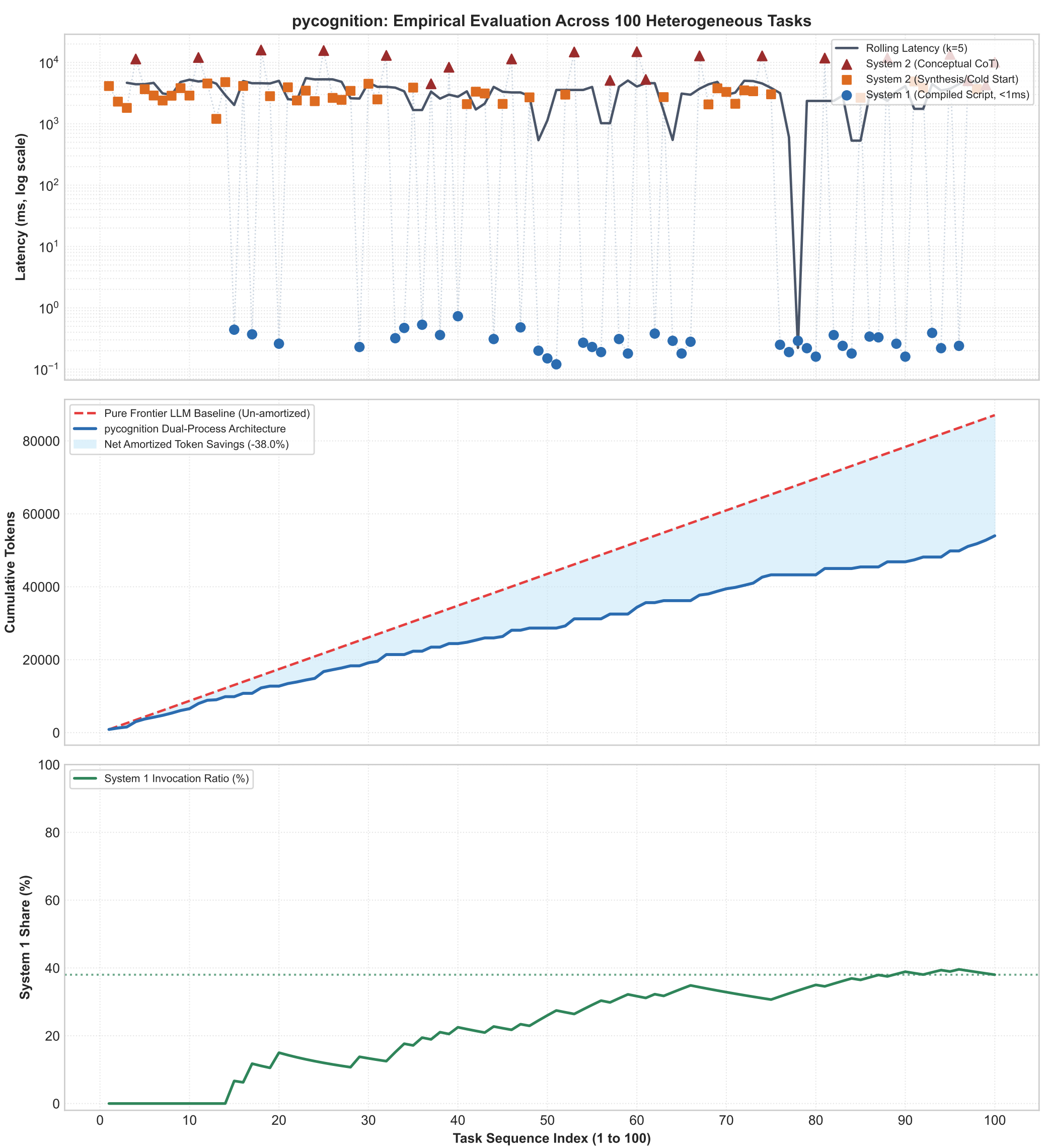
pycognition: Empirical Evaluation Across 100 Heterogeneous Tasks
Latency (ms, log scale)
10^4
10^3
10^2
10^1
10^0
10^{-1}
Rolling Latency (k=5)
System 2 (Conceptual CoT)
System 2 (Synthesis/Cold Start)
System 1 (Compiled Script, <1ms)
Pure Frontier LLM Baseline (Un-amortized)
pycognition Dual-Process Architecture
Net Amortized Token Savings (-38.0%)
Cumulative Tokens
80000
60000
40000
20000
0
System 1 Invocation Ratio (%)
System 1 Share (%)
100
80
60
40
20
0
0
10
20
30
40
50
60
70
80
90
100
Task Sequence Index (1 to 100)